

WEEK 1 (Pravalika 2520030402)

- *Install Linux VM, configure GCC, set up a Git repository and project structure, understand shell architecture, build an initial Makefile, and analyze fork()/exec() and parent-child processes.*

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        printf("fork failed\n");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execl("/bin/ls", "ls", NULL);

        printf("exec failed\n");
        exit(1);
    }
    else
    {
        printf("Parent waiting for child...\n");

        wait(NULL);
    }
}
```

Output:

```
Parent Process
Parent PID: 845
Parent waiting for child...
Child Process
Child PID: 846
Parent PID: 845
2520030001_Skill      2520030003_Skill      a.out      dirl      execsc3.c  exp2.c      p3.c      practical1.c  switch.c
2520030002_Practical 2520030402_Practical  addition   execsc.c   execsc4.c  filesystem.c p4.c      practical2.c
2520030002_Skill     2520030402_Skill     addition.c  execsc1.c  execsc5.c  'gcc addition.c -o add' p5.c      skill1.c
2520030003_Practical Forge05              addition.c.save execsc2.c  expl.c     p2.c      practical1  switch
Child process completed
```